Environ Monit Assess (2026) 198:57
https://doi.org/10.1007/s10661-025-14907-5

RESEARCH

Environmental and spatial determinants of aquatic and semi-aquatic Heteroptera diversity in Amazonian savanna streams

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Received: 26 June 2025 / Accepted: 11 December 2025
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Abstract This study investigated how environmental and spatial factors structure aquatic (Nepomorpha) and semi-aquatic (Gerromorpha) Heteroptera communities in savanna streams of Marajó Island, eastern Amazon. A total of 1,038 individuals (32 species, seven families) were collected from 22 streams exposed to varying degrees of anthropogenic disturbance. Three new geographic distribution records were documented, including one for Brazil and two for the state of Pará. Habitat integrity did not significantly influence species richness, abundance, or overall community composition. In contrast, limnological variables significantly affected the composition of Nepomorpha, whereas Gerromorpha and total Heteroptera showed no such response. Variance partitioning revealed a strong influence of environmental variables, while spatial structure had no independent effect. Our findings indicate that land-use changes act as key environmental filters, favoring tolerant generalist species and reducing overall diversity and abundance. The predominance of apterous individuals and low oxygen concentrations reflect degraded

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Published online: 20 December 2025

conditions typical of disturbed savanna streams. The low sensitivity of the habitat integrity index when applied to Amazonian ecosystems reveals the need for a specific index tailored to Amazonian savannas. These results reinforce the ecological importance and vulnerability of Amazonian savanna streams, which are often neglected in conservation planning. Aquatic Heteroptera, particularly Nepomorpha, proved to be effective bioindicators of land-use impacts, underscoring the urgency of evidence-based policies to protect these fragile and increasingly threatened freshwater systems.

Keywords Aquatic Insects · Bioindicators · Stream ecology · Anthropogenic disturbance · Species composition · Water quality

Introduction

The distribution and structure of biological communities are shaped by a combination of environmental, spatial, and biogeographic factors (Barbosa et al., 2020; Leibold et al., 2010). The metacommunity framework integrates these influences and has become relevant in ecology, particularly in understanding species distributions across landscapes (Jiang et al., 2021; Leibold et al., 2004). The metacommunity approach represents a significant advancement, by jointly evaluating these environmental, spatial, and biogeographic drivers, providing a more comprehensive understanding of community distribution patterns and their underlying mechanisms (Barbosa et al., 2020; Dias-Silva et al., 2010). In freshwater systems, community structure often reflects multiple interacting mechanisms such as environmental filtering, dispersal limitations, and stochastic processes (Borthagaray et al., 2023; Heino et al., 2015). Several environmental filters act upon species selection according to their tolerances and adaptations to local conditions. These filters may be biotic (e.g., ecological interactions) or abiotic (e.g., physical and chemical habitat attributes), determining which species are capable of colonizing and persisting over time (Leibold et al., 2004). Environments offering a greater diversity of microhabitats and higher environmental heterogeneity tend to support higher species richness (Tews et al., 2004; Stein et al., 2014). Environmental heterogeneity has a direct influence on ecological processes in lotic ecosystems, shaping community dynamics at both local and regional scales (Bakonyi et al., 2022; Brasil et al., 2020; Cunha & Juen, 2017). For example, riparian vegetation in aquatic ecosystems can provide substrates, nutrients, and shading, regulate water temperature, and filter pollutants before they enter the system (Dias-Silva et al., 2020; Franco et al., 2023).

Freshwater ecosystems sustain high levels of biodiversity and are among the most threatened ecosystems globally (Dudgeon et al., 2006). Alterations to the natural conditions of streams, especially those caused by anthropogenic activities, tend to impair aquatic communities, primarily by simplifying habitat structure reducing heterogeneity (Giehl et al., 2020; Juen et al., 2016). Also, the removal, fragmentation, or replacement of riparian vegetation, decreasing habitat connectivity, hinders species dispersal and gene flow, and increases the input of sediments and fine particulate organic matter into streams, as vegetation loss exposes soil and promotes erosion (Barbosa et al., 2020). These environmental changes affects community assembly processes, as the structure of aquatic and semi-aquatic communities results from the interplay among environmental filtering, dispersal and stochastic dynamics associated with ecological drift (Hubbel, 2001; Leibold et al., 2004). Environmental and spatial filters determine which species can colonize and persist based on their ecological tolerances and dispersal capacities (Cunha & Juen, 2017; Downes & Reich, 2008). Dispersal limitation imposed by distance, habitat isolation, or intrinsic mobility constraints, can further restrict colonization and increase similarity among nearby sites (Brito et al., 2024; Cunha et al., 2020; Ditrich, 2021). Moreover, reduced vegetation cover can elevate water temperature, accelerating the development of insect larvae and pupae and negatively affecting species sensitive to environmental change (Almeida et al., 2019). These conditions favor the establishment and persistence of tolerant generalist species, often with greater dispersal capacity (Downes & Reich, 2008; Juen et al., 2016; Moy et al., 2022).

In Amazonian savanna ecosystems, streams are frequently subject to anthropogenic pressures such as agriculture, cattle grazing, and deforestation (Gomes et al., 2021; Montag et al., 2009; Narusawa et al., 2025). These pressures reduce habitat complexity, alter limnological conditions, and compromise

ecological stability. Although less extensive than rainforest ecosystems, Amazonian savannas host high levels of endemic biodiversity and unique ecological processes, often acting as transitional zones between forest and open landscapes (Amaral et al., 2019). These regions support species adapted to seasonally dry climates and heterogeneous habitats, including various aquatic systems that are poorly represented in conservation agendas. Their dynamic hydrological regimes and natural disturbances, such as fire and flooding, contribute to ecological resilience while simultaneously increasing susceptibility to anthropogenic stressors (De Lima et al., 2005; Gomes et al., 2021; Montag et al., 2009). Similarly, the ecological dynamics of Amazonian savannas resemble those of the Brazilian Cerrado, where headwater streams are strongly influenced by seasonal climatic variation, fire regimes, and land-use changes that shape aquatic insect communities and ecosystem processes (Dias-Silva, 2010); Giehl et al., 2014; Godoy et al., 2017). In addition, parallels can be drawn with African savannas, where streams and temporary water bodies sustain diverse aquatic insect assemblages that are highly responsive to seasonal flooding and drought, and are equally vulnerable to anthropogenic pressures such as agriculture and livestock expansion (Dallas & Mosepele, 2020; Wanderi et al., 2022). These cross-biome comparisons highlight that savanna streams worldwide share hydrological variability, high exposure to disturbance, and unique biodiversity, reinforcing the importance of including them in global conservation strategies.

Monitoring aquatic ecosystems and assessing habitat integrity are essential to quantify environmental variation and distinguish the effects of anthropogenic impacts, especially on riparian zones that sustain aquatic biodiversity (Bakonyi et al., 2022; Godoy et al., 2017; Wanderi et al., 2022). Aquatic insects of the suborder Heteroptera, primarily represented in aquatic environments by the infraorders Nepomorpha and Gerromorpha, are widely used as bioindicators of ecosystem disturbances due to their sensitivity to environmental changes (Dias-Silva et al., 2010; Moy et al., 2022). Nepomorpha species inhabit the water column and often occupy highly specific ecological niches, making them particularly sensitive to environmental disturbances, even subtle ones. Gerromorpha species, on the other hand, are surface dwellers that utilize habitats associated with macrophytes and riparian vegetation structures. These insects are influenced by changes in water physical and chemical conditions, riparian vegetation (which controls resource input), and broader climatic variables (Cunha & Juen, 2017; Heino, 2009). Despite the recognized ecological importance of aquatic Heteroptera, significant knowledge gaps remain, particularly regarding their distribution, ecological requirements, and responses to environmental stressors in savanna landscapes. Many species remain undescribed (Linnaean shortfall), their geographic distributions are poorly understood (Wallacean shortfall), and their ecological roles and responses to environmental change remain largely unknown (Hutchinsonian shortfall), thereby limiting scientific progress and hindering the development of effective conservation strategies (Dias-Silva et al., 2013; Hortal et al., 2015).

Here we aimed to evaluate the effects of environmental and spatial variables on the diversity of Heteroptera, with a focus on the infraorders Nepomorpha and Gerromorpha, in Amazonian savanna streams located in the Área de Proteção Ambiental do Arquipélago do Marajó (APA Marajó). We seek to understand how local anthropogenic activities influence aquatic and semi-aquatic communities testing three main hypotheses: (i) that habitat integrity positively influences the richness and abundance of Nepomorpha and Gerromorpha, as streams with better-preserved riparian vegetation, lower anthropogenic impact, and greater structural heterogeneity are expected to offer more favorable ecological conditions for these communities. Thus, higher species richness and abundance are expected in streams with high habitat integrity; (ii) limnological variables, such as water temperature, dissolved oxygen, pH, and turbidity, significantly affect the distribution and composition of Heteroptera communities. In this case, streams with more favorable physical and chemical conditions are expected to harbor greater diversity and a distinct species composition compared to degraded environments; (iii) that local environmental variables exert a greater influence on community structure than spatial variables, given the high dispersal capacity of these taxa and the determinant role of habitat conditions in species filtering. Therefore, models incorporating only environmental variables are expected to show higher explanatory power for patterns of species composition, richness, and abundance than models based solely on spatial structure.

Materials and methods

Study area

We conducted our study in the municipalities of Salvaterra (0°45′32"S, 48°30′44"W) and Cachoeira do Arari (1°00′16"S, 48°57′27"W), both situated in the Área de Proteção Ambiental do Arquipélago do Marajó (APA Marajó), established in 1989 and covering approximately 50,000 km^2 (Pará, 1989). This archipelago constitutes the largest fluvio-marine system in the world and includes 12 municipalities near the Equator (0°57′54.63"S, 49°28′31.15"W). The Marajó mesoregion occupies a strategic geopolitical position at the mouth of the Amazon River and is bordered by the Atlantic Ocean, which imparts estuarine characteristics to some water bodies, rendering them brackish (Carvalho et al., 2019). A large portion of the territory comprises naturally flooded or seasonally inundated areas, including mangrove habitats. Although local streams are classified as freshwater, they are influenced by tidal fluctuations and exhibit low salinity levels (Correa, 2005; De Lima et al., 2005).

The prevailing climate is equatorial, hot and humid, with a mean annual temperature of approximately 30 °C, and two well-defined seasons: a dry period from June to November, and a rainy season from December to May (Carvalho et al., 2019). The vegetation of the APA Marajó is strongly influenced by local hydrological regimes and encompasses a mosaic of phytophysiognomies, including ombrophilous forest (alluvial and lowland formations), secondary forests with anthropogenic influence, pioneer formations (salt/flooded fields, mangroves, and restinga), and savanna ecosystems. Savannas are subdivided into: (i) woody savanna, composed of numerous shrubs and non-stratified tree species with basal branching and the presence of palm trees along riparian zones; and (ii) savanna park, characterized by dense shrub cover, herbaceous vegetation, and scattered tree species, as well as palms and grasses along the riparian corridors (Azevedo et al., 2000; De Lima et al., 2005; França & Martins, 2006).

The region includes extensive rural and urban areas subject to intense agro-pastoral activities, including the extraction of açaí (*Euterpe oleracea* Mart.) and buriti (*Mauritia flexuosa* L.f.), buffalo and cattle ranching, and large, dominant expanses of soybean (*Glycine max* (L.) Merr.) or rice (*Oryza sativa* L.) monocultures (Meirelles Filho, 2014; Montag et al., 2009; Narusawa et al., 2025). These land-use practices have progressively encroached upon savanna ecosystems, which dominate much of eastern Marajó Island, highlighting ecologically vulnerable and understudied conditions that make this one of the most complex regions in Brazil (Amaral et al., 2019; Meirelles Filho, 2014). Aquatic ecosystems, including savanna streams and wetlands, are often integrated into agro-pastoral systems due to their low cost of conversion and are frequently modified through the promotion of pastures, use of fire, stream channelization, and the grazing of aquatic macrophytes by livestock. These practices threaten aquatic and semi-aquatic fauna, as well as the structural integrity of local stream ecosystems (Meirelles Filho, 2014; Montag et al., 2009).

Our sampling region in the APA Marajó encompassed different vegetation formations, including savanna park and woody savanna, with physiognomies resembling those of the Brazilian Cerrado biome; flooded grasslands under strong fluvio-marine/lacustrine influence and with floristic affinity to savannas; and anthropized areas throughout the region, all influenced by human activities such as soybean (*Glycine max* (L.) Merr.), rice (*Oryza sativa* L.), and even pineapple (*Ananas comosus* (L.) Merr.) cultivation, as well as buffalo and cattle ranching. Stream substrates were heterogeneous, ranging from coarse gravel to fine silt, as visually observed. Thus, although these areas are natural, they cannot be considered pristine. Also, these municipalities lie within the direct influence zone of Belém city (Fig. 1).

Sampling and data collection

We sampled 22 streams from first to third order during the Amazonian dry season, in October and November 2022, and June 2023. This period was chosen to minimize potential seasonal effects in the region, particularly those caused by rainfall-induced flooding, which alters aquatic microhabitats by expanding stream channels, often adjacent to mangrove areas or naturally flooded fields, and to avoid surface water turbulence that would hinder the activity and persistence of semi-aquatic bugs that depend on stable water surfaces. In each stream, we delineated a continuous 150-m stretch

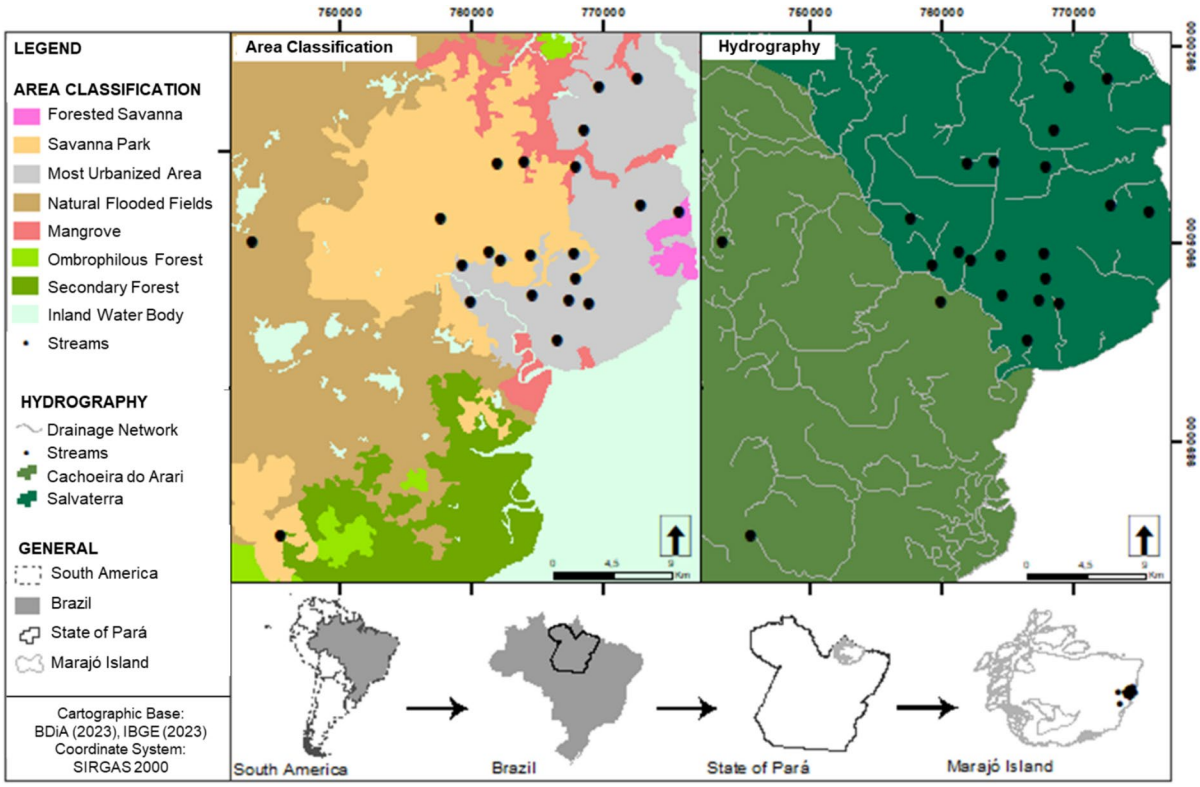

Fig. 1 Location, hydrography, and land-use classification of the streams sampled between the municipalities of Cachoeira do Arari and Salvaterra, Marajó Island, Pará, Brazil, South America

along the downstream–upstream direction and subdivided into 10 equidistant segments of 15 m each. To capture Nepomorpha (aquatic insects), a hand-held aquatic entomological net ("rapiché" type) with 0.25 mm mesh was used. Two replicate sweeps were conducted along the margins of each segment to sample the substrate, targeting a range of substrates and microhabitats. Collected material was sorted on-site for preliminary separation of aquatic insect orders (Dias-Silva et al., 2010). Specimens were fixed in 15 mL falcon tubes filled with 85% ethanol solution until laboratory processing. To collect Gerromorpha (semi-aquatic insects), active sampling was performed using a simple hand strainer, following a zigzag transect along the entire stream reach. Individuals were collected from surface populations near the margins and from observable microhabitats within the water column (Fig. 2). Samples were stored by segment in 50 mL falcon tubes containing 85% ethanol solution until further processing in the laboratory.

Specimen identification and comparison were based on regional identification keys and taxonomic descriptions for the Amazon region, Brazil, and South America, down to the lowest possible taxonomic level (Magalhães et al., 2016; Nieser, 1975). Species identification was performed using external morphology and genitalia, supported by consultation of species descriptions and comparison of specimens, always under a Leica M205 A stereomicroscope equipped with a Leica DFC 450 camera. When necessary, structures of male genitalia (parameres and phallosome) were dissected and preserved in 0.2 mL Eppendorf microtubes filled with glycerin, stored together with the corresponding adult specimens.

Environmental characterization

Habitat integrity was assessed using the Habitat Integrity Index (HII), composed of 12 items, yielding scores ranging from 0 to 1, where 1 indicates the highest level of integrity (Nessimian et al., 2008)

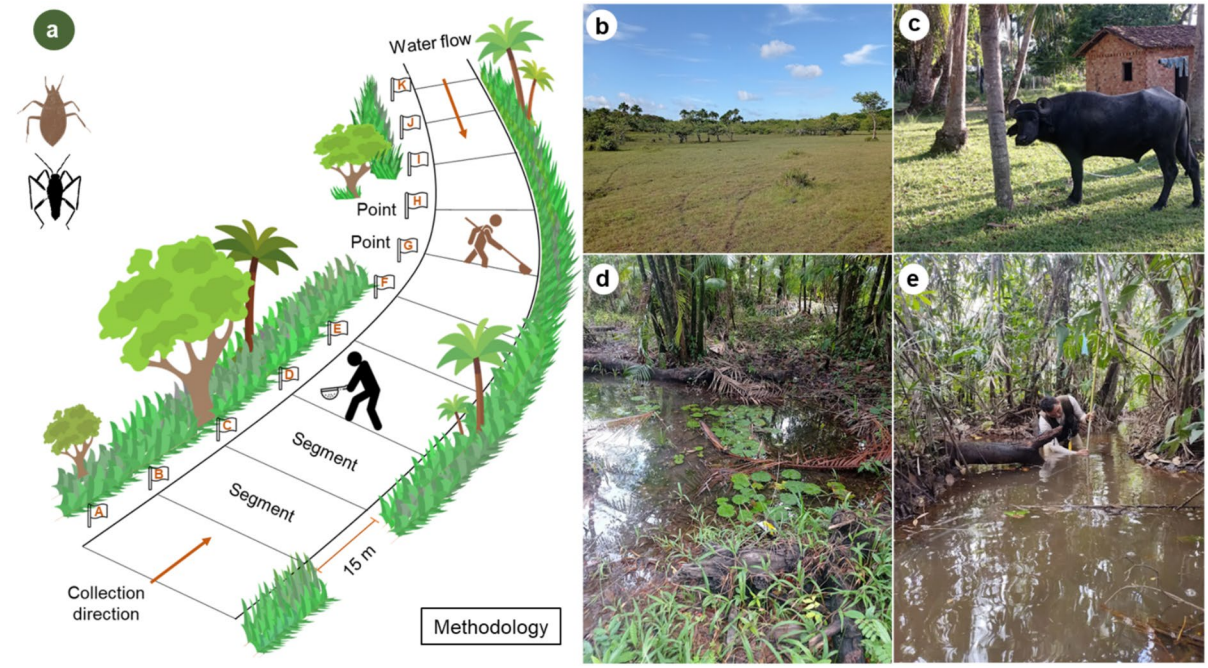

Fig. 2 Sampling scheme for Heteroptera in streams from Marajó Island. A) Sampling design, with brown indicating Nepomorpha sampling and black indicating Gerromorpha sampling; B) Marajó savanna landscape; C) Scene showing traditional household and buffalo ranching; D) Savanna stream; E) Stream under clear anthropogenic influence of pasture and pineapple, and açaí plantations

(Table S1). This index is widely used to explain the distribution of aquatic insects in Amazonian environments (Cunha & Juen, 2017).

The limnological parameters recorded and retained after multicollinearity checks were: dissolved oxygen (DO; mg/L), electrical conductivity (Cond; µS/cm), salinity (Sal; ppt), pH, and water temperature (Temp; °C) (Table S2). These parameters were measured three times per stream, at the beginning, middle, and end of the transect, and the mean value was used to represent each variable in the analyses. Measurements were obtained using a multiparameter probe AK88 (manufacturer: Akso, Brazil). Prior to the analyses, homoscedasticity, normal distribution of residuals, and absence of collinearity among variables were verified.

Environmental characterization of the streams followed the protocol proposed by Peck et al. (2006), comprising a total of 232 measured variables that encompassed channel morphology, substrate composition, hydraulic characteristics, flow type, slope, sinuosity, riparian vegetation cover, woody debris, shelter availability, and anthropogenic impact. To avoid redundancy and multicollinearity among environmental variables, we applied the Variance Inflation Factor (VIF), adopting a cutoff threshold of less than 5 (Dormann et al., 2013). Additionally, we performed a Spearman correlation among all environmental variables and retained only variables with the correlation coefficient between two variables equal to or greater than 0.5. In our selection, we also prioritized those most frequently cited in the literature on aquatic communities (Brasil et al., 2020; Juen et al., 2016). The 13 retained variables corresponded to metrics related to riparian vegetation cover, anthropogenic impact, channel morphology, substrate composition, and flow type (Table S3).

Data analysis

Each stream represented a sampling unit in this study, totaling 22 sites. We performed a Principal Coordinates Analysis (PCoA) using Hellinger-transformed species abundance data to evaluate variation in species composition and to visualize relationships between sampled communities (Gerromorpha and

Nepomorpha) in reduced-dimensional space, highlighting similarity patterns and potential groupings (Legendre & Legendre, 2012).

To summarize the variation in limnological variables and habitat integrity, we performed a Principal Component Analysis (PCA) using the variables: dissolved oxygen (DO), electrical conductivity (Cond), salinity (Sal), pH, water temperature (Temp), and the Habitat Integrity Index (HII) for each stream. The HII was included as a bubble variable to distinguish streams with higher and lower levels of habitat integrity. All variables were standardized using z-scores to account for differences in measurement scales (Legendre & Legendre, 2012). The Kaiser-Guttman criterion was applied, retaining components with eigenvalues greater than 1, i.e., those explaining more variance than expected by chance. Loadings with absolute values $\geq \pm 0.6$ were considered relevant (Grossman et al., 1991; Holubova, 2023). So, the Kaiser-Guttman criterion indicated the retention of three principal components with eigenvalues greater than 1 (PC1 = 1.92; PC2 = 1.55; PC3 = 1.09).

To test hypothesis (i), that habitat integrity positively influences the richness and abundance of Heteroptera (Gerromorpha and Nepomorpha), we conducted simple linear regressions using the richness or abundance of Heteroptera and each infraorder as response variables, and HII as the predictor variable.

To address hypothesis (ii), that limnological variables affect Heteroptera communities, we conducted a Redundancy Analysis (RDA) separately for the Heteroptera community matrix and for each infraorder, using the limnological variable matrix as the set of predictors. To reduce the influence of multiple zero values in the abundance matrices, we applied a Hellinger transformation, and predictors were selected using a forward selection procedure. Statistical significance was assessed using an ANOVA-like permutation test (Blanchet et al., 2008; Legendre & Legendre, 2012).

Finally, to address hypothesis (iii), that both environmental and spatial factors significantly influence Heteroptera (Gerromorpha and Nepomorpha), we applied a partial RDA. Environmental variables were z-score standardized, and significance was tested via permutation tests with 9,999 permutations (Legendre & Legendre, 2012). Geographic coordinates were used to compute spatial predictors based on Euclidean distance matrices. A Principal Coordinates of Neighbour Matrices (PCNM) analysis was performed using the minimum connectivity threshold as the truncation distance (Borcard et al., 2004).

All statistical analyses were conducted in the software R (version 4.3.1), using the following packages: adespatial, dplyr, ggfortify, ggplot2, ggrepel, psych, REdaS, spdep, tidyverse, and vegan (Bivand & Wong, 2018; Dray et al., 2025; Maier, 2022; Oksanen et al., 2025; Revelle, 2024; Slowikowski, 2024; Wickham, 2016; Wickham et al., 2019, 2023).

Results

Community description and stream characteristics

A total of 1,038 Heteroptera individuals were collected, distributed across seven families and 32 species. Of these, 75 individuals belonged to Nepomorpha, comprising four families, four genera, and 13 species, all of which were winged adults. The remaining 963 individuals were Gerromorpha, encompassing three families, 12 genera, and 19 species. Among these, 953 were apterous, and 10 were winged, including seven Stridulivelia tersa, two *Calivelia conata*, and one *Stridulivelia alia*. Three new geographic distribution records were identified: one for Brazil, *Ranatra adelomorpha* Nieser, 1975; and two for the state of Pará, *Belostoma denticolle* Montandon, 1903 and *Rheumatobates trinitatis* (China, 1943) (Table 1).

The Principal Coordinates Analysis (PCoA) did not reveal the formation of distinct groups (between Nepomorpha and Gerromorpha), explaining 22.82% of the total variation in species composition, 16.89% accounted for by the first axis, and 13.59% by the second (Fig. 3). Despite the lack of clear compositional grouping, the ordination revealed a tendency for Nepomorpha and Gerromorpha to occupy partially distinct environmental spaces, with considerable overlap among species. This pattern reflects high ecological plasticity, as species respond individually to environmental gradients rather than forming discrete assemblages.

The Habitat Integrity Index (HII) ranged from 0.18 to 0.67 (mean ± standard deviation), reflecting sites with low to moderate levels of environmental integrity (Table S1). The limnological variables showed wide variation among the sampled streams, reflecting the heterogeneous environmental conditions across

Table 1 List of Heteroptera species recorded in the savanna streams of Marajó Island. New records for the state are marked with "*" and for the country with "!"

Species	Abundance	Species code
NEPOMORPHA		
Belostomatidae		
Belostoma amazonum Estévez & Polhemus, 2001	2	Bel_ama
Belostoma denticolle Montandon, 1903 *	7	Bel_den
Belostoma micantulum (Stål, 1860)	1	Bel_mic
Notonectidae		
Martarega chinai Hynes, 1948	1	Mar_chi
Martarega gonostyla Truxal, 1949	1	Mar_gon
Martarega membranacea White, 1879	4	Mar_mem
Nepidae		
Ranatra adelomorpha Nieser, 1975!	2	Ran_ade
Ranatra macrophtalma Herrich-Schäffer, 1849	22	Ran_mac
Ranatra magna Kuitert, 1949	9	Ran_mag
Ranatra mediana Montandon, 1910	6	Ran_med
Ranatra obscura Montandon, 1907	18	Ran_obs
Ranatra siolii De Carlo, 1970	1	Ran_sio
Micronectidae		
Tenagobia incerta Lundblad, 1929	1	Ten_inc
GERROMORPHA		
Gerridae		
Cylindrostethus palmaris Drake e Harris, 1934	278	Cyl_pal
Limnogonus aduncus Drake e Harris, 1933	6	Lim_adu
Limnogonus recurvus Drake e Harris, 1930	5	Lim_rec
Microvelia pulchella Westwood, 1834	3	Mic_pul
Microvelia sp.1	1	Mic_13
Neogerris visendus (Drake e Harris, 1934)	17	Neo_vis
Rheumatobates minutus flavidus Drake e Harris, 1942	14	Rhe_m_fla
Rheumatobates trinitatis (China, 1943) *	5	Rhe_tri
Tachygerris adamsoni (Drake, 1942)	2	Tac_ada
Telmatometra fusca Kenaga, 1941	7	Tel_fus
Telmatometra retusa Kenaga, 1941	50	Tel_ret
Veliidae		
Callivelia conata (Hungerford, 1929)	2	Cal_con
Euvelia lata Polhemus & Polhemus, 1984	63	Euv_lat
Rhagovelia sp.1	98	Rha_salgro
Stridulivelia alia (Drake, 1957)	55	Str_ali
Stridulivelia strigosa (Hungerford, 1929)	4	Str_str
Stridulivelia tersa (Drake e Harris, 1941)	278	Str_ter
Mesoveliidae		
Mesovelia amoena Uhler, 1894	5	Mes_amo
Mesovelia mulsanti White, 1879	33	Mes_mul

sites (Table S2). In total, the environmental dataset comprised 13 variables, selected based on their ecological relevance to aquatic and semi-aquatic Heteroptera (Table S3).

The PCA explained 75.9% of the total variability in the limnological data, with 31.9% attributed to the first component (PC1), 25.9% to the second (PC2), and 18.1% to the third (PC3). The limnological

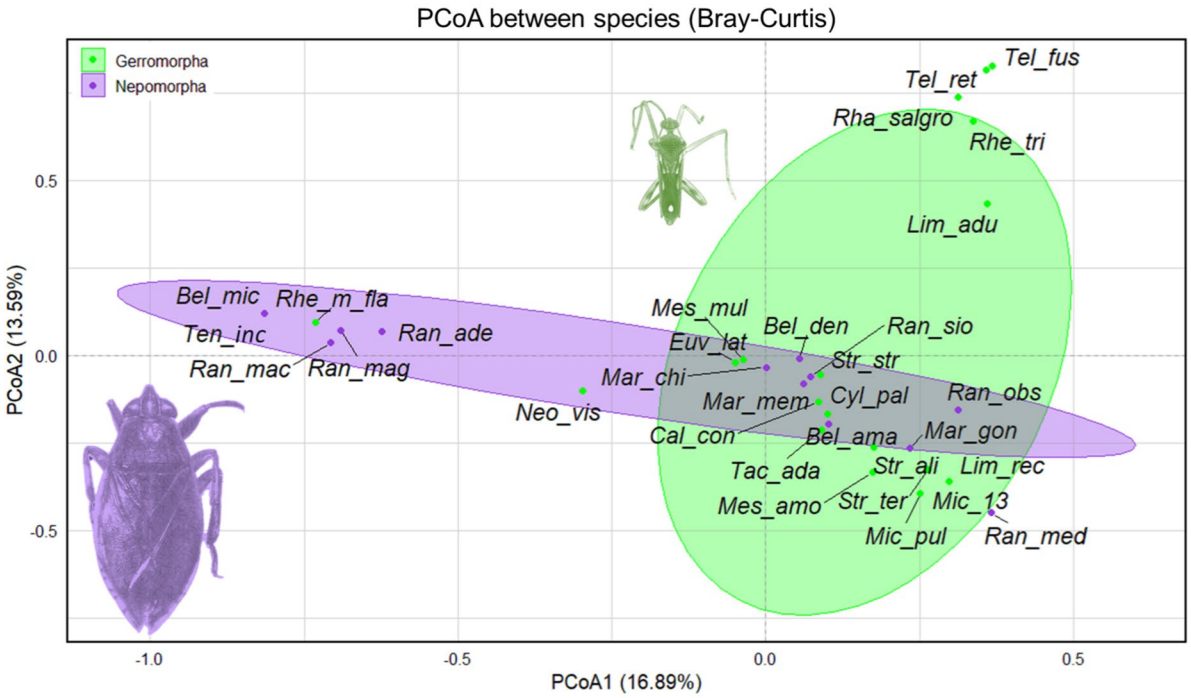

Fig. 3 Ordination of samples based on species composition by infraorder (Nepomorpha and Gerromorpha), summarized through Principal Coordinates Analysis (PCoA). Colors: Purple = Nepomorpha; Green = Gerromorpha. Species acronyms can be found in the Table 1

variables were distributed along gradients, with PC1 primarily represented by salinity with positive loading, and temperature (Temp) with a negative loading. PC2 was positively influenced by pH and conductivity, and negatively by dissolved oxygen (DO). PC3 showed positive loadings for temperature, pH, electrical conductivity, and dissolved oxygen (Fig. 4).

Suborder Heteroptera

There was no significant relationship between habitat integrity, measured by the HII and either the species richness ($R^2 = 0.018$; $p = 0.542$) or abundance ($R^2 = -0.105$; $p = 0.678$) of Heteroptera. In the RDA analysis, the limnological variables did not explain the variation in Heteroptera species composition (adjusted $R^2 = 0.000$; $F = 0.913$; $p = 0.534$).

Only PCNM2 was retained through forward selection ($F = 1.688$; $p = 0.022$). In the partial RDA, environmental variables had a significant effect (adjusted $R^2 = 0.082$; $F = 1.600$; $p < 0.001$), while spatial variables had no effect (adjusted $R^2 = 0.000$; $F = 0.407$; $p = 0.944$). The environmental variables that significantly influenced community composition were: Agricultural Human Impact per Segment – Riparian Margin, Proximity Index to Pasture, and Non-Agricultural Human Impact > 10 m from Stream. Environment and space jointly explained only 0.6% of the total variation, and 91% of the variation remained unexplained by either matrix.

Infraorder Gerromorpha

There was no significant relationship between habitat integrity and either the species richness ($R^2 = 0.050$; $p = 0.768$) or abundance ($R^2 = 0.000$; $p = 0.860$) of Gerromorpha. Likewise, in the RDA analysis, limnological variables also did not significantly explain variation in Gerromorpha species composition (adjusted $R^2 = 0.017$; $F = 0.889$; $p = 0.463$).

The spatial variable retained through forward selection during the PCNM analysis corresponded to a regional scale, in which only one vector showed positive and significant spatial autocorrelation,

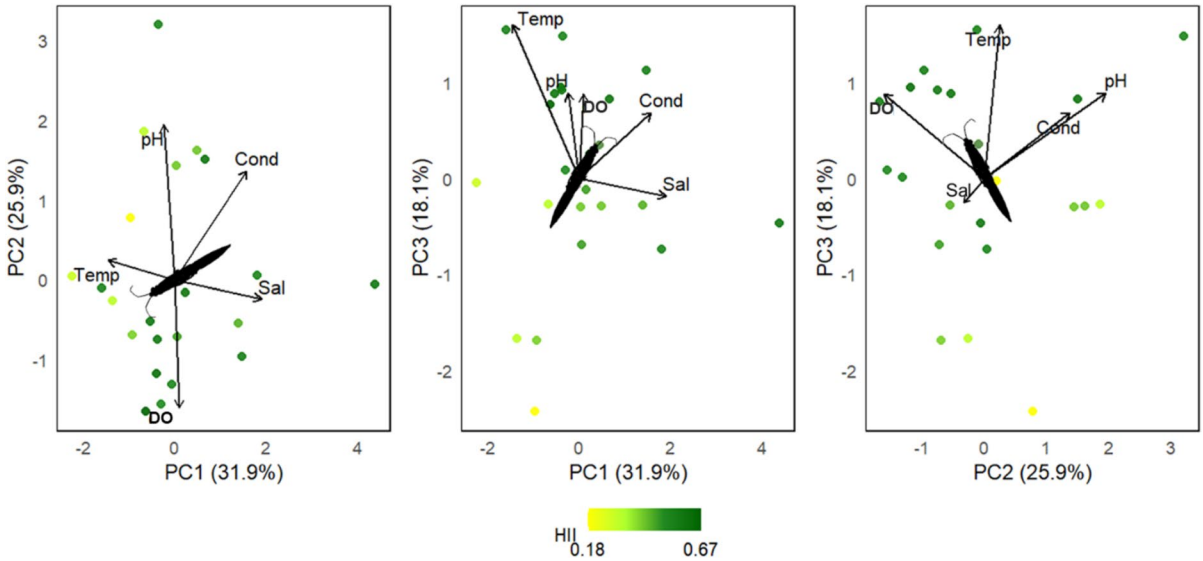

Fig. 4 Principal Component Analysis (PCA) showing the ordination of limnological variables for each stream sampled in Marajó Island and their relationship with the HII. Abbreviations: DO = Dissolved Oxygen; pH = Hydrogen Potential; Temp = Temperature; Sal = Salinity; Cond = Electrical Conductivity; HII = Habitat Integrity Index

PCNM2 (F = 1.938; p = 0.011). In the partial RDA, environmental variables had a significant effect on Gerromorpha communities (adjusted $R^2 = 0.519$; F = 2.773; $p < 0.001$), whereas spatial variables had no significant effect (adjusted $R^2 = 0.034$; F = 1.545; p = 0.196). The environmental variables that significantly influenced community composition were: Presence of Ground Vegetation per Transect, Presence of Canopy and Understory per Transect, Standard Deviation of Large Tree Canopy Cover, Proximity Index to Pasture, and Non-Agricultural Human Impact > 10 m from Stream. Environment and space jointly explained 0.6% of the total variation, while 43% remained unexplained by either matrix, indicating that these factors do not act in isolation but interact to influence the dynamics of Gerromorpha communities.

Infraorder Nepomorpha

There was no significant relationship between habitat integrity and either the species richness ($R^2 = 0.000$; p = 0.830) or abundance ($R^2 = 0.010$; p = 0.296) of Nepomorpha. On the other hand, in the RDA analysis, limnological variables explained 97.91% of the variation in Nepomorpha communities, and the relationship was statistically significant (adjusted $R^2 = 0.371$; F = 4.155; p = 0.028). The axes contributing most to the variation were RDA1 (88.34%) and RDA2 (9.57%). According to the RDA, the most influential variables were salinity, which positively affected the abundance of *Ranatra macrophtalma* and *Ranatra magna*, and dissolved oxygen, which positively influenced *Ranatra obscura* and *Belostoma denticolle* (Fig. 5).

The spatial variable retained through forward selection during the PCNM analysis corresponded to a broad spatial scale, in which only one vector (PCNM1) exhibited positive and significant spatial autocorrelation (F = 1.354; $p < 0.001$). In the partial RDA, environmental variables had a significant effect on Nepomorpha communities (adjusted $R^2 = 0.077$; F = 1.641; p = 0.043), whereas spatial variables did not show a significant effect (adjusted $R^2 = 0.016$; F = 1.254; p = 0.261). The environmental variables that significantly influenced community composition were: Agricultural Human Impact per Segment – Riparian Margin, Agricultural Human Impact per Segment > 10 m from Stream, and Mean Substrate Size. Moreover, 91% of the variation was not explained by either matrix (Fig. 6).

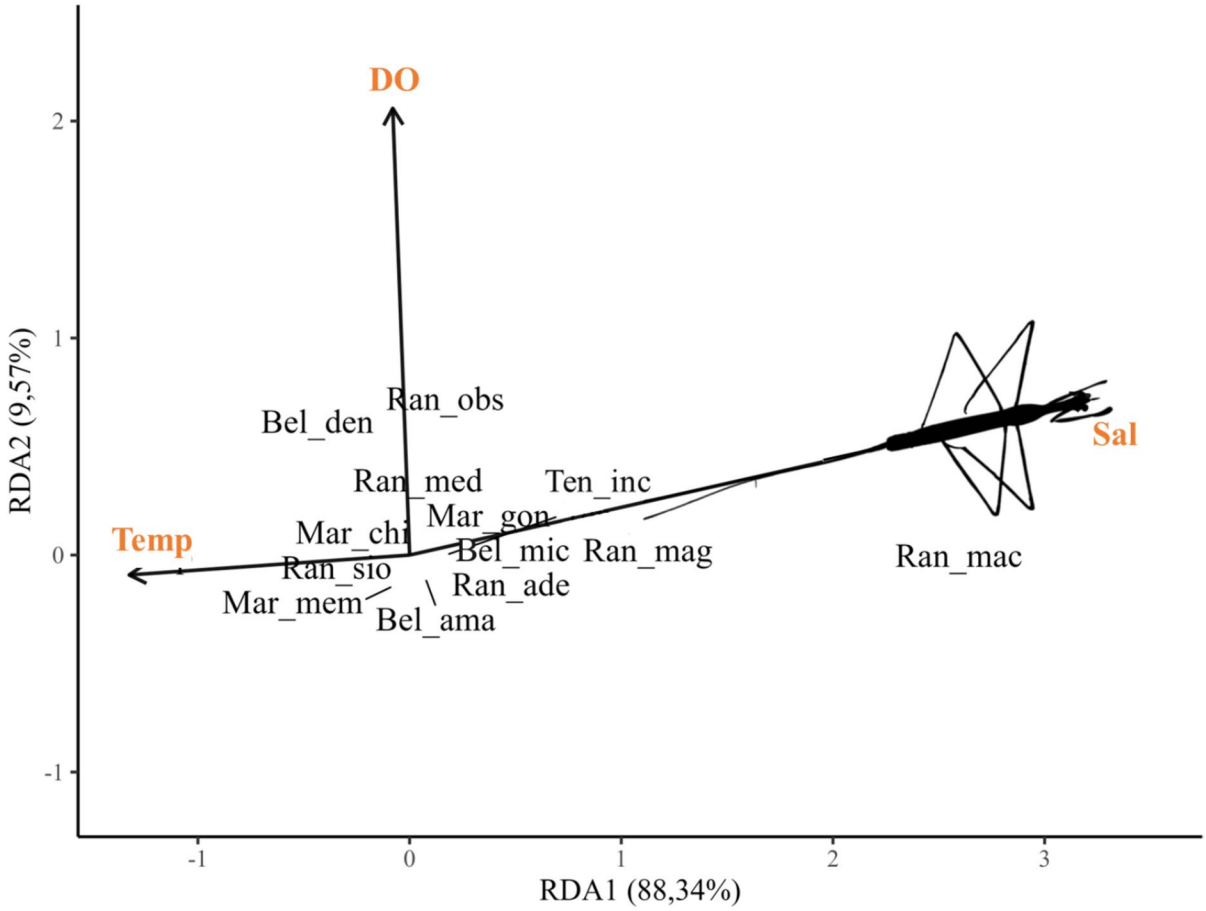

Fig. 5 Redundancy Analysis (RDA) showing the variation in Nepomorpha species composition in relation to limnological variables. Abbreviations: DO = dissolved oxygen; Temp = temperature; Sal = salinity. Species acronyms can be found in the Table 1

Discussion

Our study revealed a high diversity of species, including new records for Brazil, the Amazon region, and the state of Pará. Despite the savanna-like phytophysiognomy, the Heteroptera species composition was similar to that recorded in Amazonian forested areas (Godoy et al., 2024; Guterres et al., 2020). Notably, we report the occurrence of *Rheumatobates trinitatis*, a species typically associated with mangrove and brackish environments, previously known only from coastal regions of northeastern and southeastern Brazil (Canejo et al., 2023; Nery et al., 2025). Its occurrence in Marajoara streams may be associated with the transitional nature of these aquatic systems, which experience periodic tidal influence and slight salinity variation, creating brackish habitats. In addition, *R. trinitatis* shows high dispersal ability, which may have favored its colonization of inland freshwater environments.

Our first hypothesis, that stream habitat integrity would influence species richness and abundance, was not supported. This outcome suggests that environmental integrity, as measured by indices developed for forested systems, may have lower explanatory power for structuring Heteroptera communities in savanna streams. The methodology proposed by Nessimian et al. (2008), which represents a robust and widely applied approach for assessing habitat integrity in Amazonian forested environments, might therefore capture only partially the ecological dynamics of savanna systems, where riparian cover is naturally discontinuous, solar exposure is higher, and primary productivity is elevated. These conditions

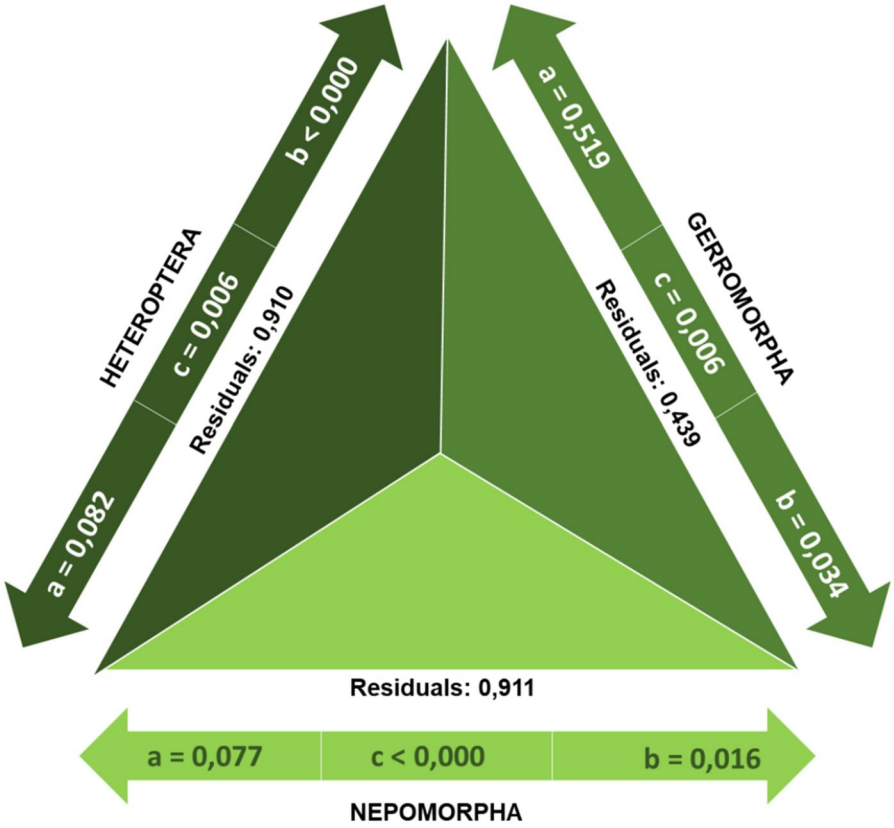

Fig. 6 Venn diagram showing the partitioning of explained variation in Heteroptera (dark green), Gerromorpha (middle green) and Nepomorpha (light green) communities between environmental and spatial components. Abbreviations: a=environmental; b=spatial; c=both

promote increased dissolved oxygen through photosynthetic activity and water turbulence, despite higher temperatures typically associated with greater light input. Although discontinuous vegetation can reduce allochthonous organic matter input, the elevated in-stream productivity compensates for this limitation, sustaining resources for aquatic fauna. Such conditions favor communities adapted to high environmental heterogeneity (De Lima et al., 2005; Montag et al., 2009). Consequently, integrity indices calibrated for forested streams may require adaptation to account for the specific ecological characteristics of Amazonian savannas, a pattern also observed in Cerrado environments (Brasil et al., 2020). Therefore, future efforts should focus on developing or calibrating habitat integrity indices specifically designed for savanna environments, allowing a more accurate assessment of ecological conditions in these unique ecosystems.

Stream environmental conditions in the region were variable, with most measured limnological variables showing signs of alteration from their natural state due to livestock and agricultural activities. None of the sampled streams could be considered pristine, as all were subject to some level of anthropogenic impact, including sewage discharge in more urbanized areas and land use associated with agriculture and cattle or buffalo ranching (Meirelles Filho, 2014). The majority of limnological values fell outside water quality standards, particularly pH (6–9) and dissolved oxygen levels (>6 mg/L), according to CONAMA, Resolution 2005. Nevertheless, these patterns show consistent variation with findings from other studies in the Marajó region (Costa et al., 2018; Marques et al., 2020). These are savanna streams under anthropogenic influence, characterized by low-lying, shrubby vegetation on shallow, acidic, and nutrient-poor soils with low water infiltration and high aluminum concentrations under a warm, seasonally dry tropical climate, where the loss of riparian vegetation reduces allochthonous input (leaf litter and detritus) and increases light penetration, promoting autotrophic production (macrophytes) (Carvalho et al., 2024; Montag et al., 2009; Wanderi et al., 2022). Consequently, land use acts as an environmental filter, often

vicariant, favoring tolerant species such as *Cylindrostethus palmaris* (Bakonyi et al., 2022; Mendes et al., 2020) and *Ranatra obscura*, which maintained active predatory behavior even under degraded stream conditions in the Marajó savanna, demonstrating ecological resilience to habitat disturbance (De Lima et al., 2024).

The increased abundance of Nepomorpha species in response to limnological variability (particularly salinity and dissolved oxygen) suggests heightened sensitivity to physical and chemical changes. This sensitivity may relate to dissolved ion availability, which is crucial for physiological processes like osmoregulation and cellular metabolism, especially during nymphal ontogenetic development (Cunha & Juen, 2017; Moy et al., 2022). Marajoara streams, naturally nutrient-poor with leached, aluminum-rich soils, exhibit low ionic concentrations, which may be temporarily altered during flood pulses (Carvalho et al., 2019; Correa, 2005). Conversely, the negative response of Gerromorpha to limnological variables suggests this infraorder is more influenced by large-scale factors like vegetation cover changes, land use modifications, hydrological regimes, or habitat connectivity.

These findings partially support our second hypothesis, as limnological variables only significantly affected aquatic communities. Water salinity typically acts as a physiological stressor that can disrupt metabolic functions, growth, and development in aquatic species, leading to malformations and mortality, a pattern documented in other aquatic insect groups (Bradley, 2008; Lambret et al., 2021). Historically, during dry-season months (June-November), reduced rainfall allows tidal influences to intensify across Marajoara ecosystems, resulting in brackish conditions (França & Martins, 2006). While brackish-adapted species may benefit from elevated productivity and reduced predator density, they still face habitat instability (Bradley, 2008). Notably, the low overall species abundance recorded in this study falls well below typical levels for Amazonian lotic systems, likely reflecting both anthropogenic stream alterations and hydrogeomorphic factors unique to Marajó Island, particularly tidal influences (Cunha & Juen, 2017; Guterres et al., 2020; Meirelles Filho, 2014).

Pollutant presence represents an understudied determinant in Heteroptera research, especially in regions with prolonged human activity like Marajó (Bakonyi et al., 2022; Montag et al., 2009). In this study, five of the environmentally significant variables directly or indirectly reflected anthropogenic impacts: trash accumulation, drainage pipes, construction, proximity to agricultural areas, and proximity to pasture areas. Proximity to farmland facilitates nutrient influx into streams with minimal vegetative buffers. Crucially, fluctuations in electrical conductivity and dissolved oxygen strongly correlate with water pollutants, particularly inorganic fertilizers from agricultural runoff (Ziemkiewicz & Northway, 2013). The chronically low dissolved oxygen levels observed, a known indicator of organic/chemical contamination, underscore the urgent need for protective measures in these vulnerable stream ecosystems and dependent communities (Meirelles Filho, 2014).

Variance partitioning revealed strong environmental control but no spatial structuring for both Heteroptera and its infraorders, confirming our third hypothesis. The explanatory power for Nepomorpha was approximately threefold higher than for Gerromorpha or total Heteroptera, demonstrating that savanna stream communities are primarily structured by local environmental filters through direct microhabitat conditioning. Such pattern is consistent with the active dispersal ability reported for several aquatic and semi-aquatic Heteroptera species, which enables them to explore and select suitable habitats rather than being passively dispersed (Cunha et al., 2020; Ditrich, 2021). This active dispersal behavior allows individuals to respond rapidly to local environmental gradients, thereby reinforcing the influence of habitat characteristics on community composition. Similar findings were reported in other studies, where environmental variables played a major role in shaping Heteroptera assemblages (Cunha & Juen, 2017; Dias-Silva et al., 2010). The Marajoara streams' natural heterogeneity stems from diverse geological origins, while their current preservation state reflects the island's geographic isolation, now increasingly compromised by expanding anthropogenic activities, causing savanna degradation, and potential habitat homogenization (Azevedo et al., 2000; De Castro et al., 2018; Dias-Silva et al., 2020; Narusawa et al., 2025). The spatial variables' non-significance indicates microhabitat occupancy depends on: (1) site-specific environmental conditions and (2) anthropogenic pressures, with the small study scale (~36 km)

and high dispersal capacity further explaining the lack of spatial patterning (Soininen, 2015).

The assembly of Heteroptera communities is jointly shaped by dispersal mechanisms and environmental heterogeneity in freshwater ecosystems (Cunha & Juen, 2017; Giehl et al., 2020; Heino, 2009). In the Marajó savanna, local and anthropogenic factors act as primary drivers of community composition, emphasizing both the sensitivity of aquatic insects to habitat modifications and the cascading effects of land-use changes on metacommunity dynamics (Cunha & Juen, 2017; Dias-Silva et al., 2010; Giehl et al., 2020; Moy et al., 2022).

Conclusions

Our study reveals that the savanna streams of the APA Marajó harbor a remarkably diverse Heteroptera fauna, including new distribution records for Brazil, the Amazon region, and Pará state. This discovery underscores the biological significance of these often-overlooked ecosystems in regional and national conservation agendas. Local environmental conditions, strongly influenced by anthropogenic land use, emerged as key drivers of community structure, demonstrating that land use acts as a potent ecological filter. The predominance of tolerant species and low overall abundance serve as clear bioindicators of severe environmental stress in these ecosystems.

These results serve as a warning for the conservation of Marajó streams, which must be treated as a priority. They are fragile habitats with unique biodiversity and are undergoing rapid transformation. Ignoring their ecological specificities, such as hydrogeomorphological dynamics and natural environmental heterogeneity, means compromising the integrity of ecosystems that may be on the verge of collapse.

Moreover, our data indicate that Heteroptera can serve as excellent bioindicators in Amazonian savanna environments, a region still lacking in-depth ecological research. Therefore, it is crucial to promote evidence-based public policies that support conservation and the continuous monitoring of these communities. Protecting the savanna streams of Marajó means preserving a natural heritage that links Amazonian biodiversity to the resilience of aquatic ecosystems in the face of environmental change. Above all, it is a commitment to the future of science, conservation, and the communities that depend on these environments for survival.

Acknowledgements ESL acknowledges the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES - Finance code 001), the Graduate Program in Ecology for doctoral funding (Grant No. 88887.841990/2023-00), and the French Government through the France Excellence BGF scholarship program (Campus France/Dossier 184202R). We thank the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for the productivity research grants KDS (311550/2023-1), LJ (304710/2019–9), LFAM (302881/2022-0), and TSM (311835/2023-6). We also appreciate the support of the projects INCT Sínteses da Biodiversidade Amazônica (CNPq/MCTIC/INCT-2022 58/2022, process 406767/2022-0), Programa de Pesquisa em Biodiversidade da Amazônia Oriental - PPBIO AmOr (process 441257/2023–2), and PELD-AmOr (process 445970/2024–3).

Author contribution E.S. de Lima: Conceptualization, Software, Methodology, Data curation, Writing – original draft, Writing – review & editing. T.M. Ramos: Samples collection, Writing – review & editing. L.V.S. Vareira: Samples collection, Writing – review & editing. J.C.G. Ortega: Methodology, Samples collection, Writing – review & editing. L.F.A. Montag: Samples collection, Writing – review & editing. T.S. Michelan: Writing – review & editing. L. Juen: Conceptualization, Resources, Validation, Writing – review & editing. K. Dias-Silva: Conceptualization, Supervision, Validation, Writing – review & editing. All the current authors have read and approved the manuscript order list, agree with the content, and submitted the manuscript.

Funding This study was supported by the project "Assessment of aquatic biodiversity in streams of the Marajó archipelago" (CNPq; Grant No. 449314/2014–2).

Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request. To ensure full reproducibility of the analyses, the metadata are also available from Msc. Eduarda Silva de Lima (eduardaflorestal@gmail.com).

Declarations Ethics approval and consent to participate.

All authors have read, understood, and complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for authors. All authors have reviewed and endorsed the submitted manuscript.

Competing interests The authors declare no competing interests.

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